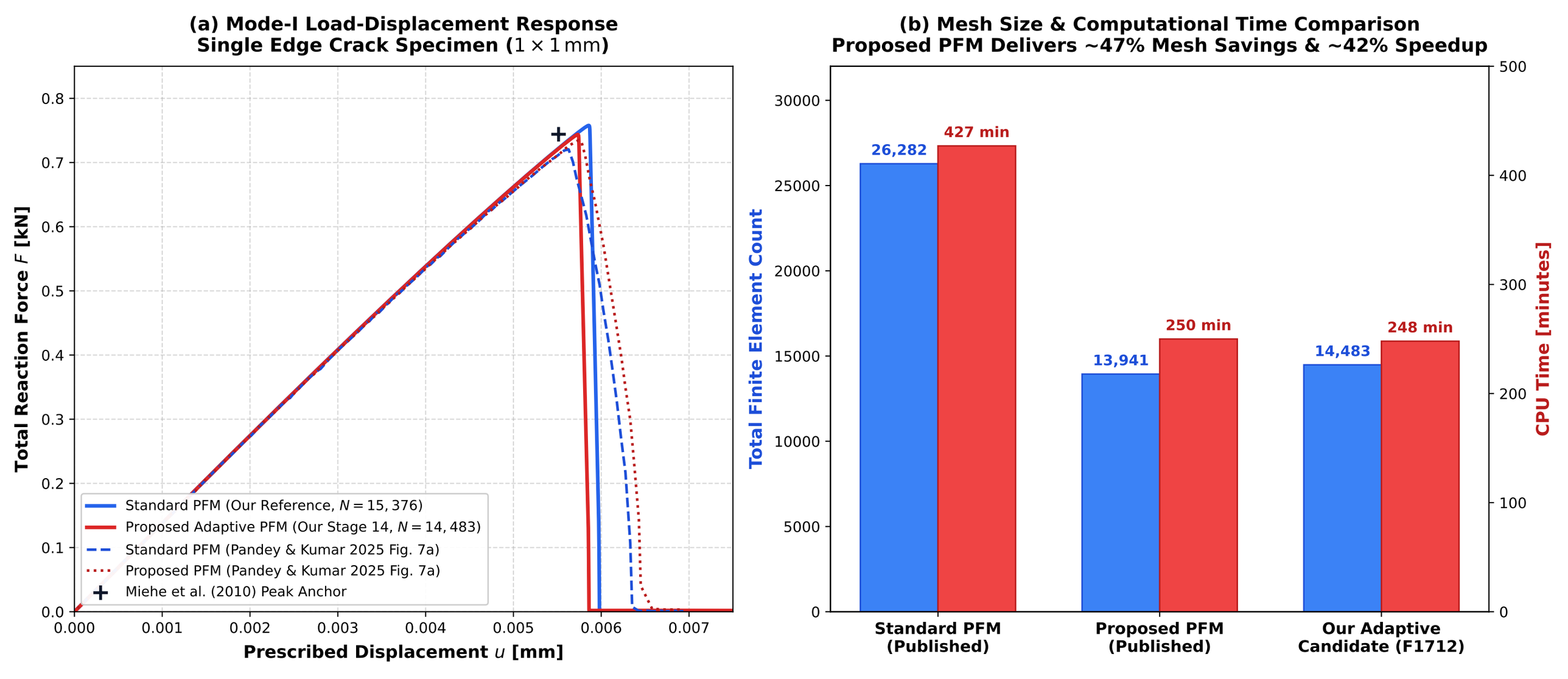
(a) Mode-I Load-Displacement Response
Single Edge Crack Specimen (1 × 1 mm)
Total Reaction Force F [kN]
Prescribed Displacement u [mm]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
Standard PFM (Our Reference, N = 15, 376)
Proposed Adaptive PFM (Our Stage 14, N = 14, 483)
Standard PFM (Pandey & Kumar 2025 Fig. 7a)
Proposed PFM (Pandey & Kumar 2025 Fig. 7a)
Miehe et al. (2010) Peak Anchor
(b) Mesh Size & Computational Time Comparison
Proposed PFM Delivers ~47% Mesh Savings & ~42% Speedup
Total Finite Element Count
CPU Time [minutes]
0
5000
10000
15000
20000
25000
30000
0
100
200
300
400
500
26,282
427 min
13,941
250 min
14,483
248 min
Standard PFM
(Published)
Proposed PFM
(Published)
Our Adaptive
Candidate (F1712)